

One Circle Equation, Three Applications

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Question: How does a driven pendulum reduce to a circle flow?

Let $m > 0$ be the mass, $L > 0$ the length, $b > 0$ the viscous damping coefficient, $g_0 > 0$ gravitational acceleration, and $\Gamma \geq 0$ a constant applied torque. Starting from

$$mL^2\ddot{\theta} + bL^2\dot{\theta} + mg_0L \sin \theta = \Gamma,$$

nondimensionalize the equation and identify the overdamped limit.

Solution. Define

$$\tau = \frac{mg_0}{bL}t, \quad \gamma = \frac{\Gamma}{mg_0L}, \quad \varepsilon = \frac{m^2g_0}{b^2L}.$$

Here primes denote differentiation with respect to τ . Since

$$\dot{\theta} = \frac{mg_0}{bL}\theta', \quad \ddot{\theta} = \left(\frac{mg_0}{bL}\right)^2\theta'',$$

division by mg_0L gives

$$\varepsilon\theta'' + \theta' + \sin \theta = \gamma.$$

The fast and slow dimensional time scales are

$$t_f = \frac{m}{b}, \quad t_s = \frac{bL}{mg_0}, \quad \varepsilon = \frac{t_f}{t_s}.$$

Consider $\varepsilon \downarrow 0$ while γ remains bounded. After a transient on the dimensional fast time scale t_f , the formal leading reduced equation is

$$\theta' = \gamma - \sin \theta.$$

More generally, if γ varies with ε , a sufficient scaling condition for this reduction is

$$\varepsilon(1 + \gamma) \longrightarrow 0,$$

or, in dimensional parameters,

$$\frac{m^2g_0}{b^2L} + \frac{m\Gamma}{b^2L^2} \longrightarrow 0.$$

To ensure that the initial layer does not produce an order-one change in θ , also assume

$$\varepsilon|\theta'(0)| \longrightarrow 0,$$

which is $m|\dot{\theta}(0)|/b \rightarrow 0$ in dimensional variables. In particular, this condition holds if $\theta'(0)$ remains bounded as $\varepsilon \downarrow 0$. The reduced equation does not retain an independently prescribed initial angular velocity.

Visualization: Driven-pendulum regimes

$0 \leq \gamma < 1$	$\gamma = 1$	$\gamma > 1$
$\sin \theta^* = \gamma$	$\theta^* = \pi/2$	no equilibrium
$\theta_s = \arcsin \gamma$	half-stable threshold	$\theta' > 0$
$\theta_u = \pi - \arcsin \gamma$	saddle-node collision	continual overturning
$\cos \theta_s > 0$	$\theta' \sim \frac{1}{2}(\theta - \pi/2)^2 \quad (\theta \rightarrow \pi/2)$	$T_\tau := \frac{2\pi}{\sqrt{\gamma^2 - 1}}$

Here θ_s is asymptotically stable and θ_u is unstable. The dimensional period for $\gamma > 1$ is

$$T_t = \frac{bL}{mg_0} T_\tau.$$

Question: What happens below the torque threshold?

Classify the equilibria of the reduced pendulum when $\gamma = 0.6$.

Solution. For $\gamma = 0.6$,

$$\theta_s = \arcsin(0.6) \approx 0.6435, \quad \theta_u = \pi - \arcsin(0.6) \approx 2.4981.$$

For $f(\theta) = 0.6 - \sin \theta$,

$$f'(\theta_s) = -0.8 < 0, \quad f'(\theta_u) = 0.8 > 0.$$

Thus θ_s is asymptotically stable and θ_u is unstable.

Question: What is the period above the torque threshold?

Compute the dimensionless period of the reduced pendulum when $\gamma = 1.25$.

Solution. Since $\gamma = 5/4$,

$$T_\tau = \frac{2\pi}{\sqrt{(5/4)^2 - 1}} = \frac{8\pi}{3}.$$

Recall: Phase difference

Let Θ be the phase of a periodic stimulus and θ the phase of an oscillator. Their **phase difference** is

$$\phi = [\Theta - \theta] \in S^1.$$

Definition: Phase locking

For a continuous autonomous phase-difference equation

$$\dot{\phi} = g(\phi),$$

the phases are **phase-locked** if $\phi(t)$ approaches an equilibrium ϕ^* of g .

Definition: Long-time frequency

Let $\tilde{\theta}(t)$ be a lift of a phase $\theta(t) \in S^1$. If the limit exists, the **long-time frequency** is

$$\bar{\omega}_\theta = \lim_{T \rightarrow \infty} \frac{\tilde{\theta}(T) - \tilde{\theta}(0)}{T}.$$

Definition: Entrainment

Two oscillators are **entrained** if their long-time frequencies exist and are equal.

Observation: Phase locking transfers a long-time frequency

Suppose $\phi(t) \rightarrow \phi^*$ and choose compatible lifts for which $\tilde{\phi} = \tilde{\Theta} - \tilde{\theta}$ converges. If either $\bar{\omega}_\Theta$ or $\bar{\omega}_\theta$ exists, then both exist and

$$\bar{\omega}_\Theta - \bar{\omega}_\theta = \lim_{T \rightarrow \infty} \frac{\tilde{\phi}(T) - \tilde{\phi}(0)}{T} = 0.$$

Thus a phase-locked pair is entrained whenever one of its long-time frequencies exists.

Definition: Phase drift

The motion exhibits **phase drift** if a lift $\tilde{\phi}(t)$ of the phase difference is unbounded as $t \rightarrow \infty$.

Definition: Phase slip

A **phase slip** occurs between times $t_1 < t_2$ if a lift of the phase difference satisfies

$$\tilde{\phi}(t_2) - \tilde{\phi}(t_1) = \pm 2\pi.$$

Question: When does a firefly lock to a stimulus?

Let

$$\dot{\Theta} = \Omega, \quad \dot{\theta} = \omega + A \sin(\Theta - \theta), \quad A > 0, \quad \Omega, \omega \in \mathbb{R}.$$

Find the phase-difference equation, the locking range, and the stable locked phase.

Solution. Define

$$\phi = \Theta - \theta, \quad \Delta = \Omega - \omega.$$

Then

$$\dot{\phi} = g(\phi), \quad g(\phi) = \Delta - A \sin \phi.$$

A fixed point exists exactly when

$$|\Delta| \leq A,$$

equivalently,

$$\omega - A \leq \Omega \leq \omega + A.$$

If $|\Delta| < A$, the stable phase is the unique solution satisfying

$$\sin \phi_s = \frac{\Delta}{A}, \quad -\frac{\pi}{2} < \phi_s < \frac{\pi}{2},$$

Question: When does a firefly lock to a stimulus? (continued)

because

$$g'(\phi_s) = -A \cos \phi_s < 0.$$

The other fixed point is unstable. If $|\Delta| = A$, the two fixed points coalesce into one half-stable, Lyapunov-unstable locked phase.

Question: How fast does the unlocked firefly drift?

For the firefly model

$$\dot{\phi} = \Delta - A \sin \phi, \quad A > 0,$$

find the drift period and mean drift rate when $|\Delta| > A$.

Solution. If $\Delta > A$, then

$$T_{\text{drift}} = \int_0^{2\pi} \frac{d\phi}{\Delta - A \sin \phi} = \frac{2\pi}{\sqrt{\Delta^2 - A^2}},$$

and

$$\bar{\omega}_\phi = \frac{2\pi}{T_{\text{drift}}} = \sqrt{\Delta^2 - A^2}.$$

For $\Delta < -A$, the same period uses $|\Delta|$ and the mean drift rate is $-\sqrt{\Delta^2 - A^2}$.

Question: What does locking look like numerically?

Let $\omega = 1$, $A = 0.4$, and $\Omega = 1.2$. Find the stable and unstable locked phases.

Solution. Here

$$\Delta = 0.2, \quad \sin \phi_s = \frac{1}{2},$$

so

$$\phi_s = \frac{\pi}{6}, \quad \phi_u = \frac{5\pi}{6}.$$

Because $\phi = \Theta - \theta$, the value $\phi_s = \pi/6$ means that the firefly phase lags the stimulus phase by $\pi/6$.

Question: What does phase drift look like numerically?

Let $\omega = 1$, $A = 0.4$, and $\Omega = 1.6$. Find the drift period and mean drift rate.

Solution. Here $\Delta = 0.6 > A$, and

$$T_{\text{drift}} = \frac{2\pi}{\sqrt{0.6^2 - 0.4^2}} = \frac{2\pi}{\sqrt{0.2}} \approx 14.0496,$$

$$\bar{\omega}_\phi = \sqrt{0.2} \approx 0.4472.$$

Proposition: Locking range for a general response function

Let $F : S^1 \rightarrow \mathbb{R}$ be continuous and let

$$\dot{\Theta} = \Omega, \quad \dot{\theta} = \omega + AF(\Theta - \theta), \quad A > 0, \quad \Omega, \omega \in \mathbb{R}.$$

Define

$$F_{\min} = \min_{S^1} F, \quad F_{\max} = \max_{S^1} F.$$

A locked phase exists if and only if

$$AF_{\min} \leq \Omega - \omega \leq AF_{\max}.$$

Proof. Since $\phi = \Theta - \theta$,

$$\dot{\phi} = g(\phi) = \Omega - \omega - AF(\phi).$$

A fixed point exists exactly when $(\Omega - \omega)/A$ belongs to the range $[F_{\min}, F_{\max}]$.

Proposition: Stability of a locked phase

Let $F : S^1 \rightarrow \mathbb{R}$ be continuous, let $A > 0$, let $\Omega, \omega \in \mathbb{R}$, and define

$$g(\phi) = \Omega - \omega - AF(\phi).$$

Assume g is locally Lipschitz, and let $g(\phi^*) = 0$. If F is differentiable at ϕ^* , then ϕ^* is locally asymptotically stable when $F'(\phi^*) > 0$ and unstable when $F'(\phi^*) < 0$.

Without differentiability, ϕ^* is locally asymptotically stable if there is a $\delta > 0$ such that

$$g(\phi) > 0 \quad (\phi^* - \delta < \phi < \phi^*), \quad g(\phi) < 0 \quad (\phi^* < \phi < \phi^* + \delta),$$

where the intervals are interpreted in a local angular coordinate on S^1 .

Proof. At a differentiable fixed point,

$$g'(\phi^*) = -AF'(\phi^*),$$

which gives the derivative test. In the second case, the phase-line arrows point toward ϕ^* from both sides.

Definition: Josephson relations

Let ϕ be the phase difference across a Josephson junction, $I_c > 0$ its critical current, I_s its supercurrent, V its voltage, e the elementary charge, and $\hbar$ the reduced Planck constant. The **Josephson relations** are

$$I_s = I_c \sin \phi, \quad V = \frac{\hbar}{2e} \dot{\phi}.$$

Definition: Shunted-junction model

For a capacitance $C \geq 0$ and resistance $R > 0$ in parallel with the junction, the **resistively and capacitively shunted-junction model** with constant bias current I is

$$C\dot{V} + \frac{V}{R} + I_c \sin \phi = I.$$

Definition: Long-time mean

For any locally integrable observable $q : [0, \infty) \rightarrow \mathbb{R}$ whose long-time mean exists, write

$$\langle q \rangle = \lim_{S \rightarrow \infty} \frac{1}{S} \int_0^S q(t) dt.$$

This number is the **long-time mean** of q .

Observation: Mean of a periodic observable

If q is periodic with period T , then

$$\langle q \rangle = \frac{1}{T} \int_0^T q(t) dt.$$

Question: What is the dimensionless junction equation?

Rewrite the shunted-junction model as an equation for ϕ and nondimensionalize it.

Solution. Substitution of the voltage relation gives

$$\frac{\hbar C}{2e} \ddot{\phi} + \frac{\hbar}{2eR} \dot{\phi} + I_c \sin \phi = I.$$

Define

$$\tau = \frac{2eI_c R}{\hbar} t, \quad i = \frac{I}{I_c}, \quad \beta = \frac{2eI_c R^2 C}{\hbar}.$$

With primes denoting τ -derivatives,

$$\beta \phi'' + \phi' + \sin \phi = i.$$

The McCumber parameter β is the dimensionless capacitance. Consider $\beta \downarrow 0$ while i remains bounded. After the initial transient,

$$\phi' = i - \sin \phi.$$

More generally, if i varies with β , a sufficient scaling condition is $\beta(1 + |i|) \rightarrow 0$. To ensure that the initial layer does not produce an order-one change in ϕ , also assume

$$\beta |\phi'(0)| \rightarrow 0,$$

which is $RC|\dot{\phi}(0)| \rightarrow 0$ in dimensional variables. In particular, this condition holds if $\phi'(0)$ remains bounded as $\beta \downarrow 0$.

Theorem: Overdamped current–voltage curve

Let $I \geq 0$, $I_c > 0$, and $R > 0$, and define

$$i = \frac{I}{I_c}, \quad \tau = \frac{2eI_c R}{\hbar} t.$$

For the overdamped junction equation

$$\phi' = i - \sin \phi, \quad V = I_c R \phi',$$

where the prime denotes differentiation with respect to τ , the long-time mean voltage is

$$\langle V \rangle = \begin{cases} 0, & 0 \leq I \leq I_c, \\ R\sqrt{I^2 - I_c^2}, & I > I_c. \end{cases}$$

Proof. Since

$$\langle V \rangle = \frac{\hbar}{2e} \langle \dot{\phi} \rangle = I_c R \langle \phi' \rangle,$$

it suffices to average ϕ' . If $0 \leq i \leq 1$, the phase approaches a fixed point, so $\langle \phi' \rangle = 0$.

If $i > 1$, the dimensionless period is

$$T = \frac{2\pi}{\sqrt{i^2 - 1}}.$$

Therefore

$$\langle \phi' \rangle = \frac{1}{T} \int_0^T \phi' d\tau = \frac{1}{T} \int_{\phi(0)}^{\phi(0)+2\pi} d\phi = \frac{2\pi}{T} = \sqrt{i^2 - 1}.$$

Hence

$$\langle V \rangle = I_c R \sqrt{i^2 - 1} = R \sqrt{I^2 - I_c^2}.$$

Question: What voltage occurs above threshold?

Let $I = 5I_c/4$, so $i = 5/4$. Compute the dimensionless period, mean angular phase rate, phase-slip event rate, and mean voltage.

Solution.

$$\begin{aligned} i &= \frac{5}{4}, & T &= \frac{2\pi}{\sqrt{25/16 - 1}} = \frac{8\pi}{3}, \\ \langle \phi' \rangle &= \frac{2\pi}{T} = \frac{3}{4}, & \nu_{\text{slip}} &= \frac{1}{T} = \frac{3}{8\pi}, \\ \langle V \rangle &= \frac{3}{4} I_c R. \end{aligned}$$

Thus the mean angular phase rate is $3/4$ radians per unit dimensionless time, while phase-slip events occur at rate $3/(8\pi)$ per unit dimensionless time.

Visualization: One equation, three interpretations

	dimensionless phase model	pendulum	Josephson junction
phase variable	ϕ	mechanical angle	quantum phase difference
phase rate	ϕ'	angular velocity	$V/(I_c R)$
drive	i	torque ratio	current ratio I/I_c
periodic term	$\sin \phi$	gravity	supercurrent
inertia	β	mass	capacitance

Fixed point $\leftrightarrow$ rest $\leftrightarrow$ zero mean voltage, while rotation $\leftrightarrow$ overturning $\leftrightarrow$ nonzero mean voltage.